Digitalization

Which of the following statements describes what sound pitch is?

Select one:

- A. The piercing sound of a whistle is an example of a low-pitched sound.
- B. The pitch is a quality of sound related to its loudness.
- C. The pitch is is an auditory sensation related to frequencies that makes it possible to judge sounds as "higher" and "lower". ✔
- D. The pitch for a sound is the equivalent to the colour for a light.

In digitisation, what is Fourier analysis useful for?

Select one:

- A. Fourier analysis is useful to replace a complex sound wave by a pure tone.
- B. Fourier analysis is useful to represent a complex waveform by a sum of simpler sine waves. ✔
- C. Fourier analysis is useful to extract low amplitude components from a complex waveform.
- D. Fourier analysis is useful to calculate the dynamic range of a complex waveform.

What is a complex sound wave?

Select one:

- A. A sound that has high frequencies is a complex sound wave.
- B. A sound that has inaudible frequencies is a complex sound wave.
- C. A sound that has multiple frequency components is a complex sound wave. ✔
- D. A sound that has a large dynamic range is a complex sound wave.

What is a signal sample?

Select one:

- A. For light, a signal sample is a wavelength at a point in space.
- B. A signal sample is the result of applying Fourier analysis to the signal.
- C. For sound, a signal sample is a frequency at a point in time.
- D. A signal sample is a signal value at a point in time and/or space. ✔

Which of the following statements describe quantisation?

Select one:

- A. Quantisation is the process or technique of obtaining a representative sample from a continuous signal.
- B. Quantisation is the process of constraining an input from a continuous or otherwise large set of values to a discrete set. ✔
- C. Quantisation is the process of mapping input values from a small set (often a discrete set) to output values in a (continuous) larger set.
- D. Quantisation is the process of approximating a set of discrete symbols by rounding their values.

What is the Nyquist rate and what is it useful for?

Select one:

- A. The Nyquist rate is the minimum rate at which a finite bandwidth signal needs to be sampled to retain all of the information. It is useful to avoid aliasing. ✔
- B. The Nyquist rate is the maximum rate at which a finite bandwidth signal needs to be sampled to retain all of the information. It is useful to avoid aliasing.
- C. The Nyquist rate is the maximum rate at which a finite bandwidth signal needs to be quantised to retain all of the information. It is useful to avoid quantisation errors.
- D. The Nyquist rate is the minimum rate at which a finite bandwidth signal needs to be quantised to retain all of the information. It is useful to avoid quantisation errors.

What are quantisation errors?

Select one:

- A. Often a quantiser supports only a limited range of possible output values and performs clipping to limit the output to this range whenever the input exceeds the supported range. The error introduced by this clipping is referred to as the quantisation error.
- B. Quantisation errors occur when the quantisation rate is too high for the signal's dynamic range.
- C. Quantisation error is the result of a low sampling rate and aliasing.
- D. Quantisation error is the difference between the analog signal and the closest available digital value at each sampling instant from the analogue to digital converter. ✔

What is aliasing?

Select one:

- A. In audio, aliasing is the result of using a low sampling rate, which translates to poor sound quality. ✔
- B. Aliasing is characterised by the lack of details in a reconstructed digital signal compared to the original analogue signal.
- C. In signal processing, aliasing is an effect that causes different signals to become indistinguishable when quantised.
- D. Aliasing occurs when a signal is sampled at a less than twice the lowest frequency present in the signal.

About the spectral power distribution of a light source, which of the following statements is not true?

Select one:

- A. Knowledge of spectral power distribution of light will enable us to describe the colour of the light.
- B. The spectral power distribution of a light, which exhibits very low energy levels for a large number of wavelengths, corresponds to white light. ✔ This is true.
- C. A light's spectral power distribution specifies the amount of power it contains at each wavelength in the visible spectrum, which lies roughly between 400 and 700 nm.
- D. A spectral power distribution is a graph of the energy levels of a light source through a range of wavelengths of light.

Color

What are metamers?

Select one:

- A. Metamers are colour stimuli that are physically distinct but that are perceived to be the same colour by an observer. ✔
- B. Metamers are colour stimuli that have identical spectral power distributions and are perceived as identical colours for a given observer.
- C. Metamers are colour stimuli that have identical spectral power distributions but different tristimulus values.
- D. Metamers are colour stimuli with identical spectral power distributions that are perceived as different colours by the human eye.

What is the role of the cone cells in vision?

Select one:

- A. Cones are responsible for colour vision in relatively bright light. They respond differently to light of different wavelengths. ✔
- B. Cones are responsible for both day and night colour vision and are most sensitive to light at wavelengths around 200 nm.
- C. Cones are responsible for vision under day light and enable us to discriminate between light and dark colours, i.e. different levels of brightness.
- D. Cones are responsible for night vision at low light levels.

What is the brightness of a colour?

Select one:

- A. Brightness is the perceived lightness or darkness of a particular colour. ✔
- B. Brightness is calculated by adding all the individual wavelengths energy of a particular colour.
- C. Brightness is related to the amount of white mixed with the colour: the more white, the brighter the colour.
- D. Brightness is related to the number of wavelengths present in a light: the more concentrated the light is around a small number of wavelengths, the brighter the colour.

Why is CMYK said to be a subtractive colour model?

Select one:

- A. CMYK is called subtractive because inks subtract the colours red, green and blue from white light. ✔
- B. CMYK is not subtractive because it is CMY (Cyan Magenta Yellow) to which K (blacK) has been added.
- C. CMYK is not subtractive because inks added on paper create many different colours.
- D. CMYK is called subtractive because inks subtract the colours cyan, magenta and yellow from white light.

In the HSV model, how is the hue represented?

Select one:

- A. By a wavelength value between 0 and 359.
- B. By a value between 0 and 255, which corresponds to the maximum number of colours that can be represented.
- C. By a value between 0 and 359, which corresponds to an angle. ✔
- D. By a name (the name of the colour).

Why is RGB said to be an additive colour model?

Select one:

- A. RGB is said to be additive because it adds red, green and blue wavelength component values to white light.
- B. RGB is not an additive colour model, it works by removing all wavelength components from the light, except the red, green and blue components.
- C. RGB is additive because the perceived colour can be predicted by summing the numeric representations of red, green and blue components. ✔
- D. RGB is not additive, it works by removing red, green and blue wavelength component values from white light.

How are S (Saturation) and V (Value) represented in the HSV model?

Select one:

- A. S is represented on the horizontal axis, and V is represented on the vertical axis of the model. ✔
- B. S is represented on the diagonal axis of the model, from the base of the cone to its surface, and V is the angle value on the circumference of the cone.
- C. S is represented on the vertical axis, and V is represented on the horizontal axis of the model.
- D. S is represented on the circumference of the cone, and V is represented on the surface of the cone.

What is the hue of a colour?

Select one:

- A. The hue is one of the main properties of a colour, which describes how "pure" the colour is.
- B. The hue is one of the main properties of a colour, typically defined using a colour name such as red, blue, green or yellow. ✔
- C. The hue is one of the main properties of a colour, it is related to the energy of the light.
- D. The hue is one of the main properties of a colour, typically characterised by a unique wavelength value.

What is the difference between CMY and CMYK?

Select one:

- A. CMY is a subtractive colour model; but CMYK is additive as it adds black as an additional colour.
- B. In CMY, black is obtained by using no ink; but in CMYK it is obtained by using the K ink.
- C. CMY is used in computer monitors; but CMYK is used in printers.
- D. CMY assumes that black can be created by mixing C, M and Y; but CMYK adds black as an additional colour. ✔

What is the tristimulus theory about? Why is it important?

Select one:

- A. The tristimulus theory explains the interactions between light, the eyes, and the brain. It is important because vision is a complex phenomenon.
- B. The tristimulus theory explains how colour vision happens in the eye. It is important because it is at the basis of several colour models. ✔
- C. The tristimulus theory explains why colour vision is subjective. It is important because vision is a complex phenomenon.
- D. The tristimulus theory explains the differences between ultra-violet, visible, and infra-red light. It is important because colour is a property of light.

Why is CMYK said to be a subtractive colour model?

Select one:

- A. CMYK is called subtractive because inks subtract the colours red, green and blue from white light. ✔
- B. CMYK is not subtractive because inks added on paper create many different colours.
- C. CMYK is not subtractive because it is CMY (Cyan Magenta Yellow) to which K (blacK) has been added.
- D. CMYK is called subtractive because inks subtract the colours cyan, magenta and yellow from white light.

What is saturation?

Select one:

- A. The saturation of a colour describes how much other colours are mixed with it: the more other colours there is, the more saturated it is.
- B. The saturation of a colour describes how much white is mixed with the colour: the more white, the more saturated is the colour.
- C. The saturation of a colour describes its purity, i.e. how concentrated it is around a small number of wavelengths. ✔
- D. The saturation of a colour describes its intensity, i.e. how much energy there is in the light.

A fully saturated colour is one with no mixture of white.
三个值中至少有一个是 0 即不含 grey-------saturated
Image

When changing an image from true-colour to index-based, what is the most likely to happen?

Select one:

- A. The image will look better because the colours will be retrieved from a standard palette of colours.
- B. Nothing will happen as the two formats are equivalent.
- C. The quality of appearance of the image is likely to decrease as some colours may not be available anymore. ✔
- D. The size of the image file will increase as it needs to store a colour look up table in addition to the pixel values.

What does it mean for an image to be index-based?

Select one:

- A. With an index-based image, colour information is retrieved from a local database.
- B. With an index-based image, colour information is encoded for some of the pixels only in order to reduce the size of the stored data.
- C. With an index-based image, colour information is not directly encoded with each pixel, but is stored in a separate colour lookup table. ✔
- D. With an index-based image, colour information is reduced to 2-bits encoding.

How are image resolution and displayed size related?

Select one:

- A. Generally, because the number of pixels is usually fixed, image resolution decreases as the displayed size increases. ✔
- B. Image resolution depends on the number of pixels stored in the image file, not on its displayed size.
- C. Image resolution increases when the image is displayed at a larger size as more pixels can be shown.
- D. Because the size of a computer display is generally fixed, image resolution only depends on the number of pixels stored in the image.

Which of the following statements defines best what image resolution is?

Select one:

- A. Image resolution refers to the quality of appearance of a digital image.
- B. Image resolution is the number of pixels per inch. ✔
- C. Image resolution is the total number of pixels in an image.
- D. Image resolution refers to the pixel dimensions expressed as the number of pixels in the horizontal and vertical directions.

About grayscale images, which of the following statements is not true?

Select one:

- A. Grayscale images only need 2 bits per pixel. ✔
- B. Grayscale images can typically display up to 256 different shades of grey.
- C. Grayscale images are typically smaller than colour images.
- D. Grayscale images are entirely sufficient for many tasks and there is often no need to use more complex colour images.

About the difference between bitmaps and vector-based images, which of the following statements is not true?

Select one:

- A. Vector-based images are subject to low resolution issues but bitmaps are not. ✔
- B. Vector-based images are limited in what they can represent, but bitmaps are not.
- C. Vector-based images can be edited easily, but bitmaps necessitate specialised software.
- D. Vector-based image files are usually much smaller than bitmaps.

What is a vector-based image?

Select one:

- A. A vector-based image is a simplified representation of a photographic image.
- B. A vector-based image is made up of points, lines, and curves that are based upon mathematical equations. ✔
- C. A vector-based image is a digitised version of a computer drawing.
- D. A vector-based image is a low resolution bitmap.

What is a bitmap image?

Select one:

- A. A bitmap is a digital image composed of a matrix of pixels. ✔
- B. A bitmap is a digital image file format with extension bmp.
- C. A bitmap is a palette of colours used for indexing digital images.
- D. A bitmap is a digital image which stores one bit per pixel.

A true-colour image is typically large. Which of the following size reduction techniques is not always possible?

Select one:

- A. Changing from true-colour to binary encoding.
- B. Changing from true-colour to grayscale.
- C. Changing from true-colour to vector-based. ✔
- D. Changing from true-colour to index-based.

About true-colour images, which of the following statements is not true?

Select one:

- A. True-colour images are stored using 24 bits per RGB colour channel. ✔
- B. In true-colour images more than 16.7 different million colours can be represented.
- C. A true-colour image stores three times more data than its grayscale version.
- D. A good quality (high pixel dimensions) true-colour image is typically a few MBytes in size.

Video and audio

Which of the following types of speech sound shows almost periodic behaviours?

Select one:

- A. Voiced ✔
- B. Unvoiced
- C. Fricative
- D. Plosive

Which of the following types of speech sound contains some silence?

Select one:

- A. Plosive ✔
- B. Fricative
- C. Voiced
- D. Vowels

MIDI has many advantages over digital audio files, but also some limitations. Which of the following statements does not describe a possible MIDI limitation?

Select one:

- A. MIDI is device dependent (e.g. playback will only be accurate if the MIDI playback device is identical to the one used for the production of the MIDI music).
- B. MIDI does not sound as realistic as real instruments, because it misses out audible natural elements such as pressing keys on the piano or strumming a guitar.
- C. MIDI can lack emotion compared to live performances.
- D. MIDI is limited to instrumental tunes and vocals, and can not be used for background sounds, jingles, etc. ✔

How are audio decibels calculated?

Select one:

- A. The sound pressure levels in decibels (dB) use a logarithmic scale and take the air pressure maximum frequency of the threshold of pain as the reference point.
- B. The sound pressure levels in decibels (dB) use a logarithmic scale and take the air pressure amplitude of the threshold of hearing as the reference point. ✔
- C. The sound pressure levels in decibels (dB) use a logarithmic scale and take the air pressure minimum frequency of the threshold of hearing as the reference point.
- D. The sound pressure levels in decibels (dB) use a logarithmic scale and take the air pressure amplitude of the threshold of pain as the reference point.

What is the normal human hearing range?

Select one:

- A. The 'normal' hearing frequency range of a healthy young person is about 20 to 20,000 Hz. Though a 'normal' audible range for loudness is from 20 to 180dB, anything over 70dB is considered damaging.
- B. The 'normal' hearing frequency range of a healthy young person is about 20 to 20,000 kHz. Though a 'normal' audible range for loudness is from 0 to 160dB, anything over 80dB is considered damaging.
- C. The 'normal' hearing frequency range of a healthy young person is about 10 to 10,000 Hz. Though a 'normal' audible range for loudness is from 60 to 160dB, anything over 130dB is considered damaging.
- D. The 'normal' hearing frequency range of a healthy young person is about 20 to 20,000 Hz. Though a 'normal' audible range for loudness is from 0 to 160dB, anything over 130dB is considered painful. ✔

What is the effect of aliasing on an audio signal?

Select one:

- A. The sampled audio signal is a good representation of the analogue audio source.
- B. The sampled audio signal misrepresents the analogue audio source. ✔
- C. The sampled audio signal is bigger (file size) than the analogue audio source.
- D. The sampled audio signal is shorter (play time) than the analogue audio source.

How can the audio signal to quantisation noise ratio (SQNR) be approximated?

Select one:

- A. It can be approximated as 6 times the signal highest frequency.
- B. It can be approximated as 6 times the signal bit depth. ✔
- C. It can be approximated as 6 times the average signal amplitude.
- D. It can be approximated as 6 times the signal noise level.

What is represented in an audio histogram?

Select one:

- A. An audio histogram depicts an estimate of the probability distribution of the signal's quantisation errors.
- B. An audio histogram depicts an estimate of the probability distribution of the signal's sampling errors.
- C. An audio histogram depicts an estimate of the probability distribution of the signal's frequencies.
- D. An audio histogram depicts an estimate of the probability distribution of the signal's amplitudes. ✔

What is meant by the "Nyquist frequency"?

Select one:

- A. The Nyquist frequency is the minimum sampling rate that satisfies the Nyquist sampling criterion for a given signal.
- B. The Nyquist frequency is twice the maximum component frequency of the signal being sampled.
- C. The Nyquist frequency is half of the sampling rate of a discrete signal. ✔
- D. The Nyquist frequency is the minimum sampling rate that can avoid audio aliasing.

What is represented in an audio spectrogram?

Select one:

- A. A spectrogram displays changes in the frequencies of a signal (Y axis) over time (X axis). Amplitude is represented on a third dimension with variable brightness or colour. ✔
- B. A spectrogram displays the signal's amplitudes (Y axis) for each frequency (X axis). Time is represented on a third dimension with variable brightness or colour.
- C. A spectrogram displays the signal's frequencies (Y axis) for each amplitude level (X axis). Time is represented on a third dimension with variable brightness or colour.
- D. A spectrogram displays changes in the amplitude of a signal (Y axis) over time (X axis). Frequencies are represented on a third dimension with variable brightness or colour.

How can an audio signal dynamic range be described?

Select one:

- A. It is the ratio of the largest sound frequency and the smallest that can be represented with a given sampling rate.
- B. It is the ratio of the largest sound frequency and the smallest that can be represented with a given bit depth.
- C. It is the ratio of the largest sound amplitude and the smallest that can be represented with a given bit depth. ✔
- D. It is the ratio of the largest sound amplitude and the smallest that can be represented with a given sampling rate.

Which of the following types of speech sound are similar to white noise?

Select one:

- A. Voiced and plosive
- B. Vowel and consonants
- C. Fricative and plosive. ✔
- D. Voiced and fricative.

About speech processing applications, only one of the following statements is correct. Which one?

Select one:

- A. With speech understanding, spoken language is translated into a different language.
- B. With speaker identification, if a speaker claims to be of a certain person, her voice is used to verify this claim.
- C. With speaker verification, the identity of a speaker is determined using her voice.
- D. With speech recognition, spoken language is translated into text. ✔

About the frame rate of a video, which of the following statements is not true?

Select one:

- A. The frame rate of a movie is typically around 24 frames per second, because this rate is similar to how we see the world.
- B. If you increase the number of frames per second the characters in the video shot will appear to be moving faster. ✔
- C. It is important to keep in mind that high frame rates increase the size of the video.
- D. In videos with low frame rates movements appear jerky, while the characters move at unnatural speeds.

Lossless compression

How does the Huffman compression algorithm work?

Select one:

- A. Huffman encoders usually create patterns (strings of information) in memory and remove repeating patterns from the data.
- B. With Huffman encoding, sequences in which the same data value occurs in many consecutive data elements are stored as a single data value and count, rather than as the original sequence.
- C. Huffman encoders usually create patterns (strings of information) in memory as data is scanned looking for repeated information.
- D. With Huffman, a variable-length code table is derived from the estimated probability or frequency of occurrence for each possible value of the source symbol. ✔

Which of the following is a correct definition of "compression rate"?

Select one:

- A. It is the ratio of the compressed sampling rate to the original rate.
- B. It is the ratio of the original file size to the size of the compressed file. ✔
- C. It is the ratio of the compressed file size to the size of the original file.
- D. It is the ratio of the original sampling rate to the compressed rate.

What is the relation between entropy and bit depth?

Select one:

- A. Entropy indicates what is the maximum number of bits required to represent a given sequence of symbols.
- B. Entropy indicates what is the average number of bits required to represent a given sequence of symbols. ✔
- C. Entropy and bit depth are not related.
- D. Entropy indicates what is the minimum number of bits required to represent a given sequence of symbols.

What type of redundancy is removed with LZW (Lempel-Ziv-Welch) encoding?

Select one:

- A. Perceptual redundancy
- B. Spatial redundancy ✔
- C. Temporal redundancy
- D. None

How does entropy encoding work?

Select one:

- A. Entropy encoding usually creates patterns (strings of information) in memory as data is scanned looking for repeated information.
- B. With entropy encoding, sequences in which the same data value occurs in many consecutive data elements are stored as a single data value and count, rather than as the original sequence.
- C. Entropy encoding compresses data by replacing each fixed-length input symbol with a variable-length output codeword. ✔
- D. Entropy encoding takes advantage of human perception by reducing the amount of data that is not perceivable.

About lossless and lossy compression, which of the following statements is not true?

Select one:

- A. Lossy compression may remove some information, whereas lossless compression only removes redundant data.
- B. Lossless compression can achieve compression of any data, whereas lossy compression depends on some data properties. ✔
- C. Lossy compression typically achieves greater compression rates than lossless compression.
- D. Lossless compression allows the original data to be perfectly reconstructed from the compressed data, whereas lossy compression permits reconstruction only of an approximation of the original data.

What type of redundancy is removed with Huffman encoding?

Select one:

- A. Perceptual redundancy
- B. Temporal redundancy
- C. Spatial redundancy
- D. Statistical redundancy ✔

Which of the following statements describes best the characteristics of lossless compression?

Select one:

- A. It is a class of data compression algorithms which permits reconstruction only of an approximation of the original data, and usually achieves great compression rates.
- B. It is a class of data compression algorithms which permits reconstruction only of an approximation of the original data, and usually achieves limited compression rates.
- C. It is a class of data compression algorithms which allows the original data to be perfectly reconstructed from the compressed data, and usually achieves limited compression rates. ✔
- D. It is a class of data compression algorithms which allows the original data to be perfectly reconstructed from the compressed data, and usually achieves great compression rates.

How does the LZW (Lempel–Ziv–Welch) compression algorithm build its dictionary of symbol patterns?

Select one:

- A. LZW works by scanning through the input data for known patterns that are already in the dictionary. When such a pattern is found, a new occurrence is added to the dictionary with the next available code. In this way, the frequency of all patterns registered in the dictionary can be calculated.
- B. LZW works by scanning through the input data for completely new patterns that are not already in the dictionary. When such a pattern is found, it is added to the dictionary with the next available code. In this way, all patterns registered in the dictionary have zero symbols in common.
- C. LZW works by scanning through the input data for known patterns that are already in the dictionary. When such a pattern is found, it is removed from the dictionary. In this way, all the patterns registered in the dictionary have a unique occurence.
- D. LZW works by scanning through the input data for successively longer patterns until it finds one that is not already in the dictionary. When such a pattern is found, it is added to the dictionary with the next available code. In this way, successively longer patterns are registered in the dictionary. ✔

What type of redundancy is removed with RLE (Run Length Encoding)?

Select one:

- A. Spatial redundancy ✔
- B. None
- C. Perceptual redundancy
- D. Coding redundancy

How do dictionary-based compression algorithms work?

Select one:

- A. Dictionary based encoders compress data by replacing each fixed-length input symbol with a variable-length output codeword.
- B. Dictionary-based compression algorithms usually create patterns (strings of information) in memory as data is scanned looking for repeated information. ✔
- C. With dictionary based encoding, a variable-length code table is derived from the estimated probability or frequency of occurrence for each possible value of the source symbol.
- D. With dictionary based encoding, sequences in which the same data value occurs in many consecutive data elements are stored as a single data value and count, rather than as the original sequence.

How does RLE (Run Length Encoding) work?

Select one:

- A. RLE compresses data by replacing each fixed-length input symbol with a variable-length output codeword.
- B. RLE usually creates patterns (strings of information) in memory as data is scanned looking for repeated information.
- C. With RLE, sequences in which the same data value occurs in many consecutive data elements are stored as a single data value and count, rather than as the original sequence. ✔ Well done.
- D. RLE takes advantage of human perception by reducing the amount of data that is not perceivable.

How can RLE (Run Length Encoding) limit the "length" to one byte?

Select one:

- A. By scanning the image file in a preprocessing step to determine the largest pixel colour value.
- B. By scanning the image file in a preprocessing step to determine the size of the largest run.
- C. By dividing the runs into blocks of no more than 256 pixels. ✔ Well done.
- D. By converting the image to grayscale.

JPEG

In JEPG, what is the purpose of the quantisation table?

Select one:

- A. To apply different amount of quantisation to different frequency components. ✔
- B. To store the colour values indexed in the image.
- C. To apply different amount of quantisation to different pixels.
- D. To store the frequency components.

In JEPG, what is DPCM (Differential Pulse Code Modulation) applied to?

Select one:

- A. The DC components. ✔
- B. The quantisation factors.
- C. The quantisation values.
- D. The AC components.

In JEPG, what is the purpose of quantisation?

Select one:

- A. Reduce the colour depth.
- B. Remove some colour information.
- C. Eliminate the high-frequency components. ✔
- D. Eliminate the low-frequency components.

About the DCT (Discrete Cosine Transform), which of the following statements is not true?

Select one:

- A. The DCT coefficients give the amplitudes of the frequency components.
- B. Most of the signal information tends to be concentrated in a few high-frequency components of the DCT. ✔
- C. The DCT is invertible.
- D. The DCT transforms pixel data from the spatial domain to the frequency domain.

With a 4:2:0 chroma sub-sampling scheme, how much compression is achieved?

Select one:

- A. 2:1 ✔
- B. 5:1
- C. 3:1
- D. 4:1

Which of the following is not a processing step in JPEG?

Select one:

- A. Colour conversion
- B. Statistical encoding ✔
- C. DCT transform
- D. Chroma sub-sampling

In digital images, what does frequency refer to?

Select one:

- A. It is the rate at which the image has been compressed.
- B. It is the rate at which the image has been quantised.
- C. It is the rate at which colour values change. ✔
- D. It is the rate at which the image has been sampled.

What is colour conversion useful for in JPEG?

Select one:

- A. Colour conversion is applied to separate colour information from luminance information. ✔
- B. Colour conversion is applied to transform the image into grayscale.
- C. Colour conversion is applied to simplify the image.
- D. Colour conversion is applied to reduce the colour depth of the image.

In JEPG, what is the zigzag reordering used for?

Select one:

- A. It is used to sort the DCT coefficients from low to high values.
- B. It is used to sort the quantised coefficients from high frequency to low frequency.
- C. It is used to make Huffman encoding more efficient.
- D. It is used to increase spatial redundancy among the quantised coefficients. ✔

About transform encoding, which of the following statements is not true?

Select one:

- A. With transform encoding, the result may not be identical to the original input, but the difference is not perceptible. ✔
- B. In transform encoding, the transformation is used to enable better (more targeted) quantisation which then results in a lower quality copy of the original input.
- C. With transform encoding, changing the representation (domain) of data can sometimes make it possible to extract details that are beyond the acuity of human perception.
- D. In transform encoding, the transformation is typically lossless (i.e. perfectly reversible).

MPEG

What is the main difference between inter frame coding and intra frame coding?

Select one:

- A. Inter frame coding exploits redundancy between successive frames and intra frame coding exploits redundancy within the frames. ✔
- B. Inter frame coding is used in JPEG, and intra frame coding is used in MPEG.
- C. Inter frame coding exploits spatial redundancy, and intra frame coding exploits temporal redundancy.
- D. Inter frame coding exploits redundancy within the frames and intra frame coding exploits redundancy between successive frames.

In MPEG, when is spatial compression used on macroblocks?

Select one:

- A. In P and B frames when a matching macroblock can be found.
- B. In I frames only.
- C. In all macroblocks of I frames and when no matching macroblock can be found in P and B frames. ✔
- D. In all frames when a matching macroblock can be found.

In MPEG, when is temporal compression used on macroblocks?

Select one:

- A. In P and B frames when a matching macroblock cannot be found.
- B. In P and B frames when a matching macroblock can be found. ✔
- C. In I frames only.
- D. In I, P and B frames when a matching macroblock can be found.

Which of the following statements describes best motion estimation?

Select one:

- A. The motion estimation process analyses previous or future frames to identify macroblocks that have changed the most.
- B. The motion estimation process analyses previous or future frames to identify macroblocks that have not or little changed. ✔
- C. The motion estimation process analyses the I frames to identify macroblocks that have not or little changed.
- D. The motion estimation process analyses all the frames of a GOP to identify macroblocks that have not changed.

In MPEG, why is the frame transmission order different from the display order?

Select one:

- A. Because the B frames need frames from the future to be decoded. ✔
- B. Because all the I frames are needed to decode an MPEG video.
- C. Because all the inter-frame coded frames need information from the future to be decoded.
- D. Because all the predicted frames need to be decoded first.

Which of the following statements about I frames is not true?

Select one:

- A. I frames are encoded using intraframe coding only.
- B. I frames can be used as reference for predicting other frames.
- C. I frames can be used for random access in an MPEG video.
- D. I frames are the most compressed MPEG frames. ✔

Which of the following statements describe best temporal redundancy and spatial redundancy?

Select one:

- A. Temporal redundancy occurs in areas of a video frame where pixel values vary by small amounts. Spatial redundancy occurs when, in a sequence of images, consecutive images contain similarities.
- B. Temporal redundancy refers to elements that are duplicated within a structure, such as pixels in a video frame. Spatial redundancy refers to pixels in two video frames that have the same values in the same location.
- C. Temporal redundancy refers to pixels in two video frames that have the same values in the same location. Spatial redundancy refers to elements that are duplicated within a structure, such as pixels in a still image. ✔
- D. Temporal redundancy is redundancy within a video frame and spatial redundancy is redundancy between video frames.

Which of the following statements about P frames is not true?

Select one:

- A. P frames are the most compressed MPEG frames. ✔
- B. P frames can be used as reference for predicting other frames.
- C. P frames can use both intraframe and interframe coding.
- D. P frames use a reference from the past for doing predictions.

In the MPEG algorithm, which of the following sequences of steps is correct?

Select one:

- A. (1) frame types are decided; (2) macroblocks are divided into 8x8 blocks; (3) blocks are preprocessed; (4) chroma subsampling is applied; (5) blocks are intra-coded or inter-coded; (6) DCT and quantisation are applied.
- B. (1) frames are preprocessed; (2) frame types are decided; (3) frames are divided into 8x8 blocks; (4) chroma subsampling is applied; (5) blocks are intra-coded or inter-coded; (6) DCT and quantisation are applied.
- C. (1) frame types are decided; (2) frames are preprocessed; (3) chroma subsampling is applied; (4) macroblocks are divided into 8x8 blocks; (5) blocks are intra-coded or inter-coded; (6) DCT and quantisation are applied.
- D. (1) frames are preprocessed; (2) chroma subsampling is applied; (3) frame types are decided; (4) macroblocks are intra-coded or inter-coded; (5) macroblocks are divided into six 8x8 blocks; (6) DCT and quantisation are applied. ✔

What is the main difference between MPEG 1 and MPEG 4?

Select one:

- A. MPEG 1 is for content encoding and MPEG 4 is for content description.
- B. MPEG 1 is for video images only. MPEG 4 includes audio through MP3 encoding.
- C. MPEG 1 is recognised as a standard, but not MPEG 4.
- D. MPEG 1 is designed for encoding entire frames, but MPEG 4 encodes discrete media objects. ✔

What is the main difference between MPEG 1 and MPEG 2?

Select one:

- A. MPEG 1 is recognised as a standard, but not MPEG 2.
- B. MPEG 1 is for video images only. MPEG 2 includes audio through MP3 encoding.
- C. MPEG 1 is designed for encoding videos in the bitrate of a Compact Disc. MPEG 2 is for broadcast-quality television. ✔
- D. MPEG 1 is for content encoding and MPEG 2 is for content description.

In MPEG, what is meant by error term?

Select one:

- A. The error term is the difference between an I frame and a predicted frame.
- B. The error term is a 16x16 block of pixel differences between two matching macroblocks. ✔
- C. The error term is the difference between the reference frame from the past and the one from the future.
- D. The error term is the frame difference between a predicted frame and its reference.

In MPEG, what do motion vectors represent?

Select one:

- A. Motion vectors represent the moving speed of animated objects in video frames.
- B. Motion vectors represent motion information between macroblocks within a video frame.
- C. Motion vectors represent the moving direction of animated objects in video frames.
- D. Motion vectors represent motion information between macroblocks in different video frames. ✔

Flag question

What are the advantages of VBR (Variable BitRate) over CBR (Constant BitRate)?

Select one:

- A. VBR encodes the video data more accurately, with more bits used in less demanding passages and fewer bits used in difficult-to-encode passages.
- B. VBR videos are much smaller than CBR videos.
- C. VBR efficiently compress the video, varying between low and high bitrates depending on the complexity of the data at a given moment. ✔
- D. VBR efficiently compresses the video, using higher bitrates than CBR.

Which of the following statements about B frames is not true?

Select one:

- A. B frames can use both intraframe and interframe coding.
- B. B frames are the most compressed MPEG frames.
- C. B frames can be used as reference for predicting other frames. ✔
- D. B frames use references from the past and from the future for doing predictions.

What is a GOP typically made of?

Select one:

- A. One I frame, one P frame and one B frame.
- B. One I frame, one P frame and several B frames.
- C. One I frame and several P and B frames. ✔
- D. Several I, P and B frames.

Perceptual encoding

Why is knowledge of the critical bands important in audio processing?

Select one:

- A. Because the critical bands reveal which sound level the ear is able to perceive at a given frequency.
- B. Because the critical bands reveal how the ear is able to discriminate (tell the difference) between different frequencies. ✔
- C. Because the critical bands reveal how the ear is able to discriminate (tell the difference) between different sound levels.
- D. Because the critical bands reveal which frequencies the ear is able to perceive at a given sound level.

In perceptual encoding, what is the purpose of subband filtering?

Select one:

- A. Using a psychoacoustic model, subband filtering allows to eliminate entire bands if they are above the threshold of hearing.
- B. Using a psychoacoustic model, subband filtering allows to determine the amount of masking for each band. ✔
- C. Using a psychoacoustic model, subband filtering allows to separate loud tones from softer tones.
- D. Using a psychoacoustic model, subband filtering allows to move from the time domain to the frequency domain.

How does AAC (Advanced Audio Coding) improves on MP3?

Select one:

- A. AAC does not improve on MP3. MP3 is better.
- B. AAC uses very different encoding techniques. It improves on MP3 by compressing more efficiently but incurs more quality loss.
- C. AAC improves on MP3 by doing lossless compression.
- D. AAC uses similar encoding techniques but improves on MP3 by offering more sampling rates and arbitrary bit rates. ✔

What does masking affect the threshold of hearing?

Select one:

- A. Masking causes the threshold of hearing to be raised across widely separated critical bands.
- B. Masking causes the threshold of hearing to be lowered, especially within a critical band.
- C. Masking causes the threshold of hearing to be raised, especially within a critical band. ✔
- D. Masking causes the threshold of hearing to disappear within a critical band.

In MP3, what are scaling factors and quantisation used for?

Select one:

- A. Non-uniform quantisation, combined with scaling factors, are used to reduce the precision (i.e. reduce quantisation errors) for bands with high Signal-to-Mask Ratios (SMRs).
- B. Non-uniform quantisation, combined with scaling factors, are used to increase the precision (i.e. reduce quantisation errors) for bands with low Signal-to-Mask Ratios (SMRs).
- C. Non-uniform quantisation, combined with scaling factors, are used to increase the precision (i.e. increase quantisation errors) for bands with high Signal-to-Mask Ratios (SMRs).
- D. Non-uniform quantisation, combined with scaling factors, are used to reduce the precision (i.e. increase quantisation errors) for bands with low Signal-to-Mask Ratios (SMRs). ✔

When does temporal masking occur?

Select one:

- A. Temporal masking occurs when a loud tone masks a softer tone of similar or higher frequency.
- B. Temporal masking occurs when a loud tone masks a softer tone of similar or lower frequency.
- C. Temporal masking occurs when after a loud tone stops, there is a small delay before a softer tone can be heard. ✔
- D. Temporal masking occurs when after a loud tone stops, there is a small delay before a higher frequency tone can be heard.

When does frequency masking occur?

Select one:

- A. Frequency masking occurs when after a loud tone stops, there is a small delay before a higher frequency tone can be heard.
- B. Frequency masking occurs when a loud tone masks a softer tone of similar or higher frequency. ✔
- C. Frequency masking occurs when after a loud tone stops, there is a small delay before a softer tone can be heard.
- D. Frequency masking occurs when a loud tone masks a softer tone of similar or lower frequency.

What is non-linear companding?

Select one:

- A. It is a lossless audio compression technique which uses less bits to encode low amplitude sound samples.
- B. It is a lossy audio compression technique which uses less bits to encode high amplitude sound samples. ✔
- C. It is an audio compression technique where high frequency audio samples are encoded using less bits.
- D. It is an audio compression technique where high frequency audio samples are encoded using more bits.

What is the threshold of hearing?

Select one:

- A. It is the minimal frequency (in hertz) at which sound can be heard according to sound level.
- B. It is the maximum frequency (in hertz) at which sound can be heard according to sound level.
- C. It is the minimal level (in decibels) at which sound can be heard according to frequencies. ✔
- D. It is the maximum level (in decibels) at which sound can be perceived without feeling pain.

What is psychoacoustics?

Select one:

- A. It is the empirical study of how the human ears and brain perceive sound. ✔
- B. It is the mathematical study of how the human ears process musical sound.
- C. It is the empirical study of how the human ears and brain process speech sound.
- D. It is the mathematical study of how the brain eliminates noise perception.

What is the main principle for achieving lossless audio compression in FLAC?

Select one:

- A. FLAC uses compressing and expanding for achieving lossless compression.
- B. FLAC uses non-linear prediction, i.e. companding.
- C. FLAC uses linear prediction, i.e. it encodes the difference between the predictor and the actual sample data. ✔
- D. FLAC does lossless compression by linearly reducing the bit depth.

What is the basic principle of perceptual encoding?

Select one:

- A. Based on psychoacoustics, perceptual encoding encodes low level sounds using less bits.
- B. Based on psychoacoustics, perceptual encoding discards sounds that cannot be perceived very well. ✔
- C. Based on psychoacoustics, perceptual encoding encodes masked sounds using more bits.
- D. Based on psychoacoustics, perceptual encoding encodes high frequency sounds using less bits.

What is the principle of ABR (Average Bit Rate)?

Select one:

- A. With ABR, the bitrate is controlled interactively by the user to remain near an average value to prioritise file size over quality.
- B. With ABR, the bitrate is maintained constant to an average value chosen by the user, for more consistent quality and file size.
- C. With ABR, the bitrate is allowed to vary for more consistent quality,but is controlled to remain near an average value chosen by the user, for a predictable file size. ✔
- D. With ABR, the bitrate is allowed to vary for more predictable file size, but is controlled to remain near an average value chosen by the user, for better quality.

What information is contained in a SMR (Signal-to-Mask Ratio)?

Select one:

- A. A SMR indicates the ratio between the frequency of a masking tone and the amplitude of the minimum nearby masked frequency.
- B. A SMR indicates the ratio between the amplitude of a masking tone and the amplitude of the minimum nearby masked frequency. ✔
- C. A SMR indicates the ratio between the frequency of a masking tone and the frequency of the nearby masked tones.
- D. A SMR indicates the ratio between the amplitude of a masking tone and the frequency of the nearby masked tones.

What are typical MP3 bitrates?

Select one:

- A. MP3 was designed to encode data at 320 kbit/s or more. A bitrate of 640 kbit/s is commonly used.
- B. MP3 was designed to encode data at 320 kbit/s or less. A bitrate of 128 kbit/s is commonly used. ✔
- C. MP3 was designed to encode data at 320 kbit/s or less, but a bitrate of 320 kbit/s is most commonly used.
- D. MP3 was designed to encode data at 320 kbit/s or more, but a bitrate of 320 kbit/s is most commonly used.